

SMART INSOLE SYSTEM

A Major Project Report Submitted to

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY

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In partial fulfillment of the requirement for the award of degree of

BACHELOR OF TECHNOLOGY

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

G. PULLA REDDY ENGINEERING COLLEGE (AUTONOMOUS): KURNOOL

(Accredited by NBA of AICTE and NAAC of UGC with A grade)

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(Accredited by NBA of AICTE and NAAC of UGC with A grade)

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PROJECT GUIDE

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DECLARATION

We hereby declare that the major project titled “**SMART INSOLE SYSTEM**” is an authenticated work carried out by us as the students of **G. PULLA REDDY ENGINEERING COLLEGE** (Autonomous), Kurnool, during 2025-2026 and has not been submitted elsewhere for the award of any degree or diploma in part or in full to any institute.

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ABSTRACT

The Smart Insole project is a wearable health-monitoring solution designed to fit inside a regular shoe and provide valuable insights about a person's walking behavior, body balance, and foot health. It works silently in the background as the person walks, stands, or performs daily activities, without requiring any extra effort or noticeable change in routine. This system is particularly helpful for individuals with diabetes, the elderly, and those recovering from injuries or surgeries. By analyzing how pressure is distributed across the foot, the smart insole can identify abnormal weight patterns, detect poor posture, and recognize signs of fatigue or imbalance. One of its key features is the ability to monitor how evenly the person walks and whether any part of the foot is under constant strain - a condition that, if ignored, could lead to serious health issues like foot ulcers or joint stress.

In addition to analyzing movement, the insole can also monitor foot temperature. A sudden or prolonged rise in temperature in a specific region of the foot could signal inflammation or early signs of infection. If the insole detects any unusual conditions such as excessive pressure, abnormal gait patterns, or temperature fluctuations it sends a real-time alert to a mobile application. This allows the user to take immediate action or consult a healthcare provider before the condition worsens.

By combining comfort, real-time monitoring, and intelligent feedback, the Smart Insole empowers users to take control of their health through everyday activities like walking, promotes early detection of problems, supports rehabilitation, encourages safe mobility, and offers a preventive care tool that is easy to use, affordable, and highly practical for daily life.

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CHAPTER 1

INTRODUCTION

1. INTRODUCTION

The **Smart Insole System** is an innovative wearable technology designed to monitor and analyze foot-related parameters in real time by embedding electronic components within a shoe insole. With the rapid advancement of embedded systems and Internet of Things (IoT) technologies, smart wearable devices have gained significant importance in healthcare, sports, and rehabilitation domains. The smart insole represents one such development that enables continuous and non-invasive monitoring of human gait and foot pressure distribution.

This system typically consists of pressure sensors, accelerometer, gyroscope and temperature sensor with a microcontroller unit integrated into a compact and flexible insole. These sensors collect data related to walking patterns, weight distribution, temperature and movement. The acquired data is processed and shared to external devices such as smartphones or computers for visualization and analysis.

The primary objective of this project is to design and develop a cost effective, reliable and user-friendly smart insole capable of real-time monitoring and data analysis. By leveraging sensor technology and wireless communication, the system aims to enhance health monitoring, improve mobility assessment, and contribute to overall well-being.

1.1 BACKGROUND

The integration of wearable technology with healthcare systems has significantly transformed the way physiological parameters are monitored. Traditional healthcare approaches often rely on periodic clinical assessments, which may not provide continuous insight into a patient's condition. This limitation has led to the development of smart wearable devices capable of real-time monitoring and data analysis.

Foot health is a critical aspect of overall well-being, particularly for individuals suffering from chronic conditions such as inflammation. Similarly, elderly individuals face challenges related to balance and stability, increasing the risk of falls and injuries. Early detection and continuous monitoring are therefore essential in preventing such issues.

Gait analysis, which involves studying the pattern of walking, is widely used in medical diagnostics, sports science, and rehabilitation. Conventional gait analysis systems are typically expensive, bulky, and confined to laboratory environments, making them impractical for everyday use. This has created a demand for portable and cost-effective solutions that can operate in real-life conditions.

The Smart Insole System emerges as a solution to these challenges by combining sensor technology, embedded systems and wireless communication. By placing pressure sensors and motion sensors inside a shoe insole, the system can continuously track foot pressure distribution, walking patterns, and temperature. The collected data can then be analyzed to detect abnormalities and provide meaningful insights.

With advancements in microcontrollers, low-power electronics, and IoT platforms, it has become feasible to design compact, efficient, and affordable smart insoles. These systems not only enhance healthcare monitoring but also support applications in fitness tracking, sports performance optimization, and rehabilitation.

Thus, the Smart Insole System represents a significant step toward accessible and continuous health monitoring, bridging the gap between clinical assessment and daily life activity tracking.

1.2 PROBLEM STATEMENT

Foot-related health issues such as abnormal gait patterns, uneven pressure distribution, and balance instability are common but often go unnoticed until they become severe. These conditions can lead to serious complications including chronic pain, injuries, and mobility limitations. Elderly individuals are more prone to falls due to reduced balance and stability, posing a significant threat to their safety and independence.

Conventional methods used to assess foot health and gait patterns are typically conducted in clinical or laboratory environments using specialized equipment. These systems are often expensive, bulky, and not suitable for continuous monitoring in daily life. As a result, real-time tracking and early detection of abnormalities remain a major challenge.

Additionally, there is a lack of accessible and user-friendly solutions that can provide continuous feedback on foot pressure and walking behavior. Without proper monitoring, individuals are unable to take preventive measures at the right time.

Therefore, there is a critical need for a portable, cost-effective, and real-time monitoring system that can be easily integrated into daily life. The Smart Insole System is proposed to address these challenges by enabling continuous monitoring of foot pressure and gait patterns, thereby supporting early diagnosis, prevention of complications, and improved overall health management.

1.3 MAIN OBJECTIVE

The primary objective of this project is to design, develop, and implement a Smart Insole System capable of continuously monitoring foot pressure distribution, gait patterns, and user movement in real time. The system aims to integrate multiple sensors within a flexible and comfortable insole structure, ensuring that the user can wear it for extended periods without discomfort or disruption to daily activities.

Another key objective is to accurately capture and process data related to pressure variations across different regions of the foot. This data is used to analyze walking behavior, detect abnormalities such as uneven weight distribution, and identify potential health risks at an early stage. The system is also intended to support wireless communication, enabling seamless transmission of data to external devices such as smartphones or computers for real-time visualization and analysis.

Furthermore, the project focuses on developing a cost-effective and energy-efficient solution that can be widely accessible, especially for individuals requiring continuous monitoring, such as elderly people and athletes. The system also aims to provide meaningful feedback to users, helping them to know overall foot health.

In addition, the Smart Insole System is designed to be scalable and adaptable for various applications, including healthcare monitoring, sports performance analysis, and rehabilitation support. By combining embedded systems, sensor technology, and IoT-based communication, the project seeks to create a reliable and practical solution for foot health monitoring challenges.

CHAPTER 2

LITERATURE REVIEW

2. LITERATURE REVIEW OF RELATED PROJECTS

1. Benny Chen et al. (2015) – Smart Insole for Gait Analysis:

Benny Chen and his team developed a smart system using force-sensitive resistors (FSRs) to measure plantar pressure distribution. Their project demonstrated the effectiveness of real-time gait monitoring in identifying walking abnormalities and supporting rehabilitation.

2. Feng Lin et al. (2016):

Feng Lin, Aosen Wang, Yan Zhuang, and Machiko R. Tomita developed a smart insole system for unobtrusive gait monitoring in daily life. Their work focused on replacing traditional laboratory-based gait analysis with a wearable solution capable of continuous real-time monitoring.

3. Hwang et al. (2017) - Wearable Smart Insole System:

Hwang and his research team designed a wearable smart insole integrated with pressure sensors and wireless communication modules. The system was capable of collecting and transmitting real-time data for gait analysis and activity recognition. Their work emphasized portability and user comfort.

4. Rajesh Singh (2018) – Embedded System-Based Monitoring:

Rajesh Singh focused on the use of microcontroller-based embedded systems for real-time applications. His work demonstrated how Arduino and ATmega microcontrollers can be used to interface sensors and process data efficiently, supporting the development of cost-effective smart insole systems.

The proposed Smart Insole System introduces several advancements over prior works. Unlike existing systems, it incorporates activity mode differentiation separately calibrating thresholds for walking and running gaits significantly improving diagnostic accuracy. The system features multi-user profile management with session history storage, enabling longitudinal health tracking and comparative analysis across multiple sessions, which no referenced study supported. A real-time IoT cloud dashboard built on Blynk with live gait waveform visualization, tri-zone pressure mapping, and foot temperature monitoring provides a level of real-time observability absent in previous works. The inclusion of both timer-based and free-mode analysis gives flexibility suited to both clinical and daily-use scenarios. Most distinctly, the system generates professional reports that not only assess the current session but also compare it against the user's historical data identifying trends, flagging pressure risks, and delivering personalized rehabilitation recommendations a capability not present in any of the four referenced studies.

CHAPTER 3

PROJECT METHODOLOGY

3. PROJECT METHODOLOGY

3.1 BLOCK DIAGRAM OF SMART INSOLE SYSTEM

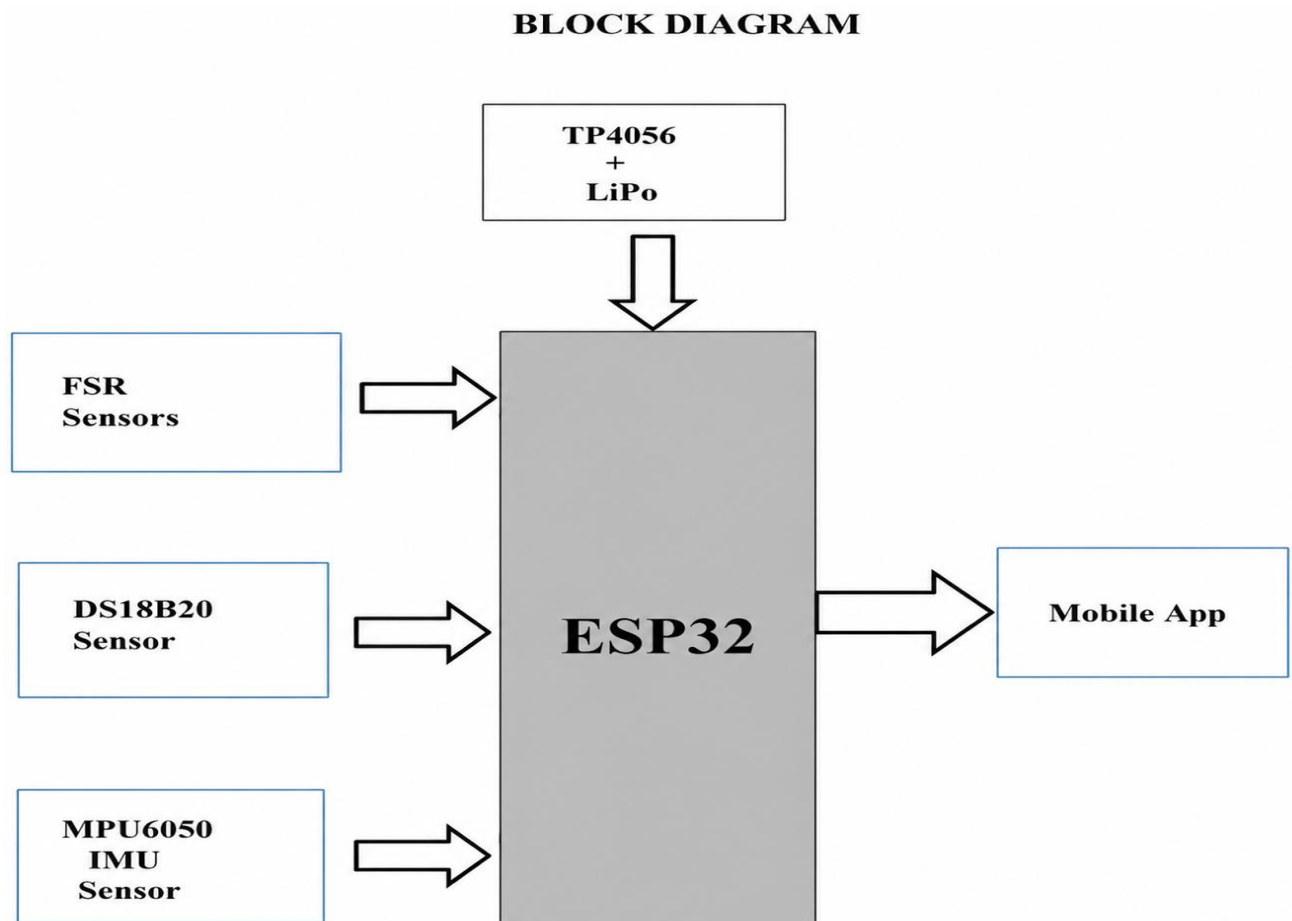


Fig 3.1 Block diagram of Smart Insole System

The above figure shows the block diagram of “Smart Insole System”, consists of ESP32 microcontroller as the central unit, powered by a LiPo battery with a TP4056 charging module. The system includes FSR sensors to measure foot pressure, a DS18B20 sensor to monitor temperature, and an MPU6050 IMU sensor to detect motion and gait patterns.

3.1.1 METHODOLOGY

The Smart Insole System is developed by integrating sensors such as FSR, MPU6050, and DS18B20 with an ESP32 microcontroller to monitor foot parameters. The system is powered by a Li-Po battery supported by a TP4056 charging module for efficient operation. These sensors continuously collect data related to foot pressure, motion, gait pattern, and temperature during user activity. The Collected data is processed by the ESP32 to analyze walking behavior and identify any abnormalities. After processing, the data is shared through Wi-Fi to mobile application. The real- time data is visualized using platforms like Blynk IoT for easy monitoring by the use.

3.2 CIRCUIT DIAGRAM OF SMART INSOLE SYSTEM

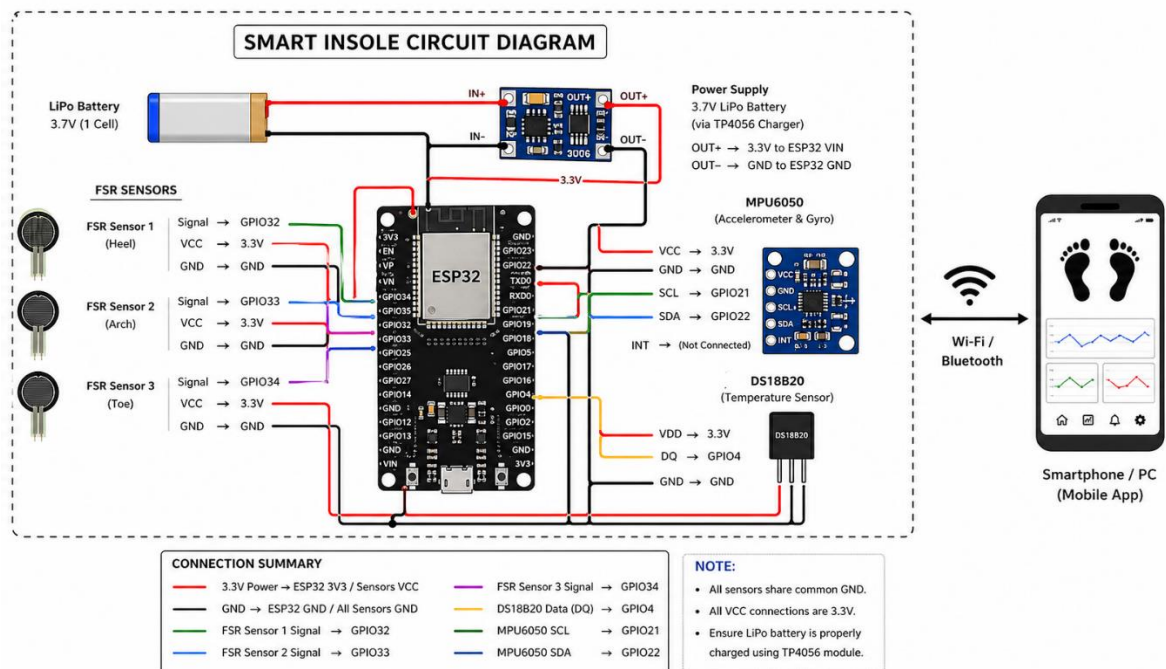


Fig 3.2 Circuit Diagram of Smart Insole System

3.2.1 CIRCUIT DESCRIPTION

The Smart Insole System circuit is designed around the ESP32 microcontroller, which acts as the central control and processing unit. It manages data collection from all connected sensors and performs the required computations for monitoring foot health and movement. The ESP32 is chosen because of its compact size, low power consumption, and ability to interface with both analog and digital sensors. In this system, it receives inputs from pressure, motion, and temperature sensors, processes the information, and supports overall device operation in real time.

The power supply section consists of a 3.7V LiPo battery connected through a TP4056 charging module. This arrangement ensures safe battery charging and provides a stable power source for the circuit. The TP4056 protects the battery from overcharging and manages the charging cycle efficiently. Its output is connected to the ESP32 power input and ground, enabling portable and rechargeable operation. This makes the Smart Insole System suitable for continuous use as a wearable device without the need for constant external power.

For sensing pressure, three FSR sensors are connected to the analog-capable pins GPIO32, GPIO33, and GPIO34 of the ESP32. These sensors are placed at important foot regions such as the heel, arch, and toe to measure pressure distribution. The resistance of each FSR changes according to the force applied, and this variation is read by the ESP32 as analog signals. In addition, the MPU6050 motion sensor is connected using the I2C protocol, with SCL linked to GPIO21 and SDA to GPIO22. This sensor captures acceleration and angular movement, enabling gait and balance analysis.

The DS18B20 temperature sensor is connected to GPIO4 as its data line, with a 4.7k ohm pull-up resistor for stable communication. It continuously monitors foot temperature and provides digital readings to the ESP32. All sensors share a common ground and operate using the 3.3V supply line. The circuit is structured to ensure reliable sensing, efficient data handling, and seamless integration of multiple parameters. This hardware configuration supports accurate real-time monitoring and forms the foundation of the Smart Insole System.

3.3 FLOW CHART OF SMART INSOLE SYSTEM

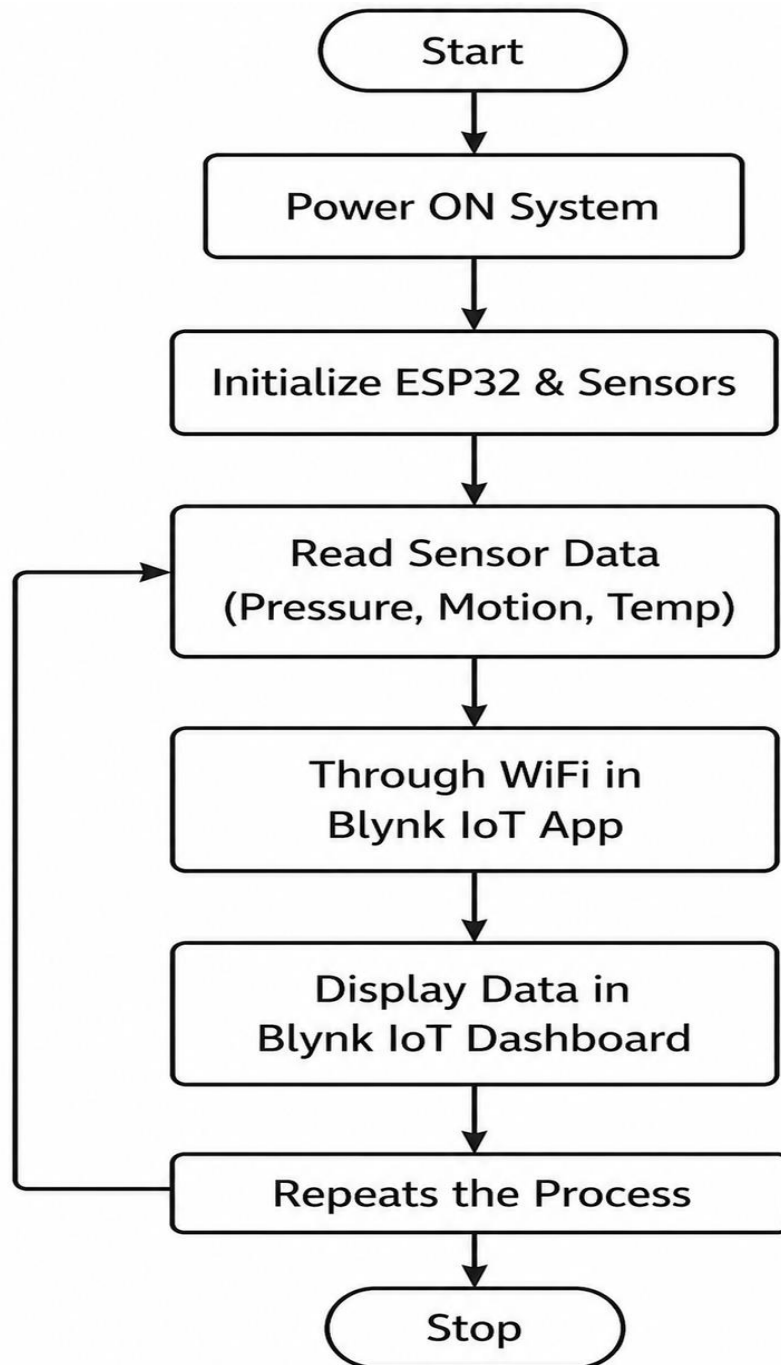


Fig 3.3 Flow Chart of Smart Insole System

The above figure shows the flowchart begins with powering the system and initializing all sensors. The sensors continuously collect data, which is processed and analyzed by the ESP32. The processed data is then transmitted wirelessly and displayed to the user. This cycle repeats continuously for real-time monitoring.

3.3.1 WORKING PRINCIPLE OF INSOLE SYSTEM

Phase 1: Setup and Initialization

The process begins the moment the user powers on the device. Inside the shoe, the "brain" of the system (the microcontroller) wakes up and performs a "health check" on all its connected sensors.

At this stage, the system is waiting for instructions from the mobile app. Before starting, the user or a caregiver decides how long the monitoring session should last—for example, a 30-second walking test or a 2-minute activity tracking session. This timer is crucial because it defines the window of time during which the system will gather and calculate the averages of your movement.

Phase 2: The Active Monitoring Session

Once the user presses "Start," the system enters a high-speed data collection mode. This is where the magic of "Sensor Fusion" happens—where different sensors work together to tell a complete story of human movement. At the same time, the MPU6050 sensor measures acceleration and angular motion. This helps in analyzing gait pattern, posture, and movement stability. By monitoring motion across multiple axes, the system can detect irregular walking behavior or sudden imbalance. The DS18B20 sensor continuously measures foot temperature, adding another important parameter for overall condition monitoring.

1. Capturing Pressure (Weight Distribution)

Three specialized pressure sensors are positioned at the Heel, Midfoot, and Toe. As you walk, these sensors act like electronic scales, measuring exactly how much of your body weight is pressing down on each part of the foot.

The Heel: Measures the initial impact of your step.

The Midfoot: Detects if your arch is collapsing or if you have a "flat-foot" walking style.

The Toe: Measures the force of your "push-off" to move forward.

2. Tracking Motion (Gait Validation)

While the pressure sensors feel the weight, a motion sensor (accelerometer) measures the "jerk" and "tilt" of the foot.

The "Jerk" Test: The system is programmed to only count a step if it feels a specific vertical vibration. If you are just standing and shifting your weight, the pressure sensors might change, but the motion sensor tells the brain, "Wait, the foot hasn't actually moved forward," so it won't record a false step.

Forward Intent: It specifically looks for a forward motion before deciding that a physical step has occurred.

3. Monitoring Heat (Plantar Temperature)

At the same time, a digital thermometer inside the insole checks the temperature of your skin every few seconds. In daily life, our feet stay within a very specific temperature range. If a part of the foot begins to heat up unnaturally, it is often the very first sign of trouble—even before the person feels any pain.

Phase 3: Data Synthesis and Reporting

As the countdown timer runs, the system isn't just showing live numbers; it is memorizing your performance. It adds up all the pressure readings and all the temperature readings taken during that 30 or 60-second window.

1. The Closing Calculation

The moment the timer hits zero, the system stops sampling. It immediately calculates the Average Pressure for the heel, middle, and toe, and the Average Temperature for the entire session. It also calculates your Cadence (how many steps you took per minute).

2. Clinical Comparison (Why it matters)

This is the most important part for human health. The system compares your averages against Normal Human Benchmarks.

Walking Speed: If you walk slower than 100 steps per minute, the system recognizes a "Cautious" or "Impaired" gait, which is a sign of aging or injury.

Flat Foot Warning: If your mid-foot pressure is too high, it warns you of a "Flat Foot Trend," which can cause long-term pain.

Infection Alert: If your average foot temperature is above 38°C, it flags "Abnormal Plantar Heat," telling you to check for inflammation or infection.

CHAPTER 4

ESP32

4. ESP32 Board

4.1 ESP32 DESCRIPTION

The ESP32 acts as the main control unit (brain) of the Smart Insole System. It is responsible for collecting, processing, and transmitting data from the sensors embedded inside the insole. In this system, multiple pressure sensors are placed at key points of the foot (heel, toe, arch). These sensors continuously measure the pressure distribution while a person walks or stands. The ESP32 reads these analog signals through its built-in ADC (Analog-to-Digital Converter).

After collecting the data, the ESP32 processes it to analyze walking patterns, balance, and weight distribution. It can detect abnormalities such as uneven pressure, improper gait, or excessive load on specific areas of the foot. One of the most important roles of the ESP32 is wireless communication. Using its built-in Wi-Fi or Bluetooth, it sends the processed data to a mobile application or cloud platform in real time. This allows users, doctors, or caregivers to monitor foot health remotely.

The ESP32 is also energy-efficient, which makes it suitable for wearable devices like insoles. It can operate in low-power modes to conserve battery while still maintaining continuous monitoring.

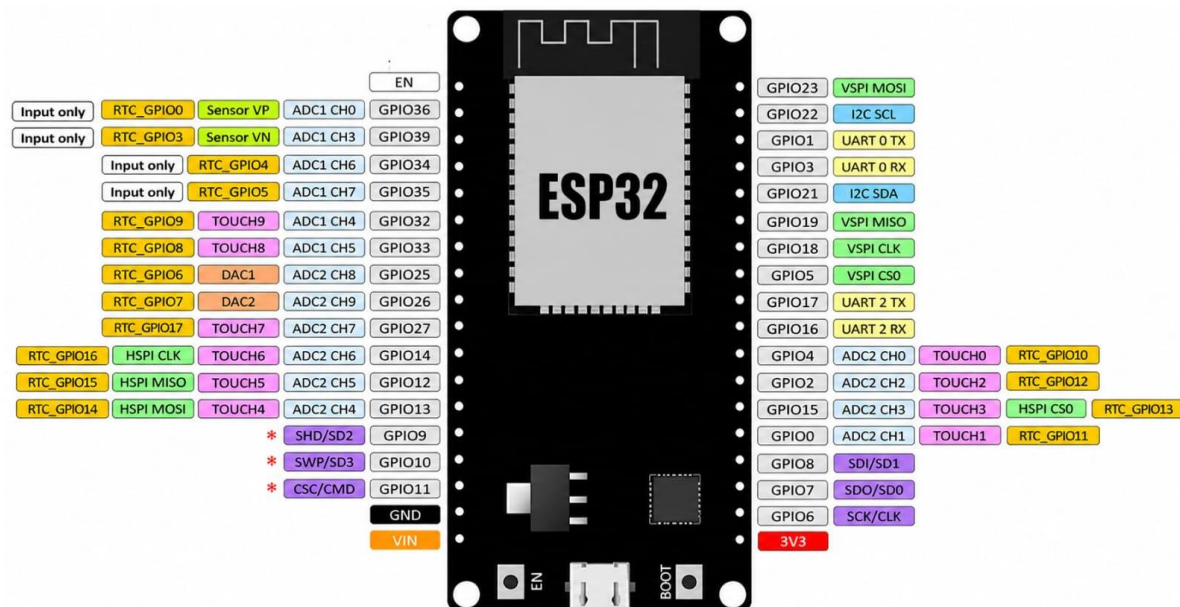


Fig 4.1: ESP32 Board.

4.2 ESP32 PIN DIAGRAM DESCRIPTION

Vin: This pin is used to provide power to the ESP32 board from an external source (typically 5V to 12V). If powered via USB, this pin outputs ~5V.

3.3V: This is the regulated output pin from the onboard voltage regulator. It provides power to sensors and components.

GND: This pin is used to ground the ESP32 board and complete the circuit with your components.

EN (Enable): This is the Reset pin. Connecting it to Ground will restart the ESP32.

ADC (Analog to Digital Converter): 18 channels (GPIO32-39, 0, 2, 4, 12-15, 25-27) for reading analog sensors (0-3.3V, 0-4095). ADC1 is recommended for reliable reading.

Digital Pins (GPIO 0–33): These pins are used as digital input or output pins. Most can be used to turn things on/off or read buttons, except for a few "strapping" pins used during boot.

Touch Pins: Specific GPIOs (like 4, 13, 14, 27, 32, 33) that can sense the electrical charge of a human touch without a physical button.

I2C Pins (GPIO 21, 22): Used for communication with external devices like OLED displays or high-precision sensors (SDA is GPIO 21, SCL is GPIO 22).

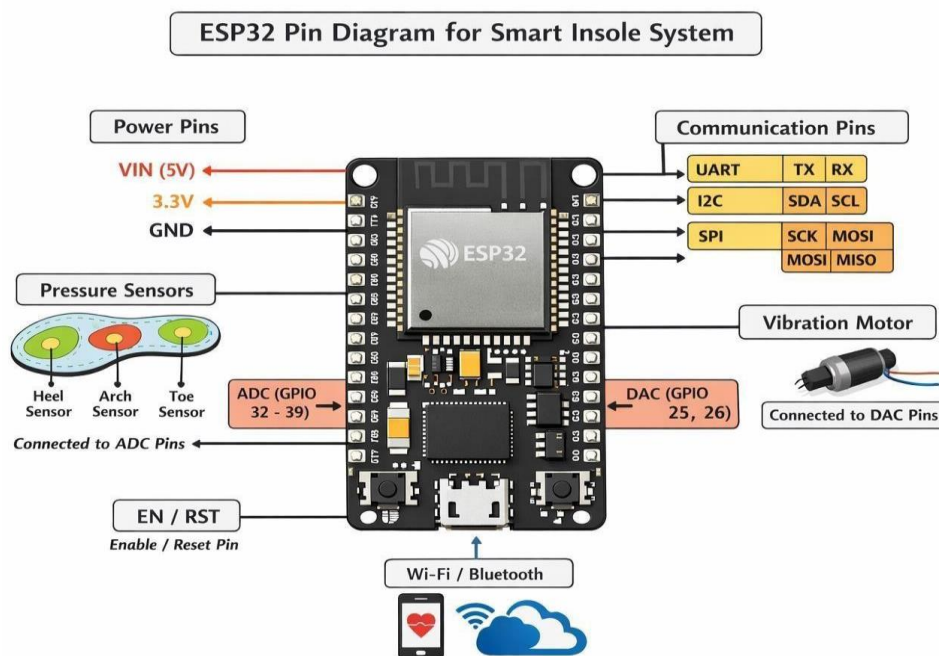


Fig 4.2 ESP32 Pin diagram

4.2.1 ESP32 SPECIFICATIONS :

Microcontroller	ESP32
Operating Voltage	3.3V
Input Voltage(recommended)	5V-9V
Input Voltage(limit)	6V-12V
Digital I/O Pins	25 TO 28
PWM Pins	16 CHANNELS
Analog Input Pins	12 TO 15 PINS (12 BIT RESOLUTION)
Communication protocol	UART x 3, SPI x 3, I2C x 2, CAN BUS, I2S
DC Current per I/O Pin	20 mA
DC Current for 3.3V Pin	600 mA to 1000 mA
USB Interface	Mini-USB
Flash Memory	4 MB

TABLE 4.2.1: ESP32 Specifications

CHAPTER 5

**DESCRIPTION OF ON BOARD HARDWARE
COMPONENTS**

5. DESCRIPTION OF ON BOARD HARDWARE COMPONENTS

5.1 Power Supply Li-ion / Li-Po Battery

The Li-Po (Lithium Polymer) battery is used as the primary power source for the Smart Insole System. It provides a stable and continuous supply of power to all the components including sensors and the ESP32 microcontroller. The Battery is lightweight and compact, making it ideal for wearable applications like smart insoles. It supports portability, allowing the system to operate without any external power source. The battery has a high energy density, which ensures longer operating time. It is rechargeable, making it cost-effective and convenient for regular use. The battery ensures uninterrupted functioning of the system during daily activities. It also supports efficient power management when combined with charging modules. Proper usage of the battery enhances the reliability of the system. Thus, the Li-Po battery plays a vital role in powering the device.



Fig 5.1: LI-PO BATTERY

5.2 DS18B20 TEMPERATURE SENSOR

The DS18B20 temperature sensor measures temperature using semiconductor-based sensing technology. Inside the sensor, silicon semiconductor junctions are used because their electrical properties change with variations in temperature. As the surrounding temperature increases or decreases, the voltage across these semiconductor junctions changes in a predictable way. The sensor contains an internal analog-to-digital converter that transforms this analog voltage variation into digital data. It then sends this data directly to the ESP32 through a one-wire communication interface.

This internal semiconductor structure makes the sensor highly accurate and reliable. In the Smart Insole project, it continuously monitors foot temperature, helping to identify unusual heat changes that may indicate inflammation, infection, or circulation problems.



FIG 5.2 DS18B20 SENSOR

5.3 FORCE SENSITIVE RESISTOR (FSR) SENSOR

The FSR (Force Sensitive Resistor) sensor detects pressure through a special pressure-sensitive conductive material placed between two flexible layers. Internally, it contains thin polyester sheets with conductive traces and a semi-conductive polymer layer made of carbon-based particles. When no force is applied, the particles remain loosely connected, resulting in high electrical resistance. As external pressure is exerted, the layers compress, bringing the conductive particles closer together and creating more electrical pathways. This reduces the resistance of the sensor. The resistance change is then converted into a voltage signal, which is read by the ESP32 microcontroller. In the Smart Insole System, this allows the sensor to measure foot pressure distribution accurately at different points such as the heel, arch, and toe.



Fig 5.3 FSR SENSOR

5.4 MPU6050 SENSOR

The MPU6050 sensor works on MEMS (Micro-Electro-Mechanical Systems) technology, which combines tiny mechanical structures with electronic circuits on a silicon chip. Internally, it contains microscopic silicon-based proof masses, capacitive plates, and vibrating structures. For acceleration sensing, a suspended proof mass shifts slightly when motion occurs, causing a change in capacitance between internal plates. This capacitance variation is converted into electrical signals that represent movement along different axes. For rotational sensing, the gyroscope section uses vibrating microstructures that respond to angular motion through Coriolis force. These internal mechanical changes are processed into digital values for acceleration and angular velocity. In the Smart Insole System, this enables accurate monitoring of walking patterns, posture, balance, and sudden falls, making it essential for gait analysis and mobility tracking.



Fig 5.4 MPU6050

5.5 TP4056 CHARGING MODULE

The TP4056 is a battery charging module used for safely charging the Li-Po battery in the Smart Insole System. It provides a constant current and constant voltage charging method to ensure efficient battery charging. The module includes protection features such as overcharge, over-discharge, and short circuit protection. It helps in maintaining the safety and longevity of the battery. The TP4056 module allows charging through a USB interface, making it easy and convenient to use. It ensures that the battery is charged properly without damaging it. The module also provides stable output voltage to the system during operation. It plays an important role in managing power supply efficiently. The charging process is simple and user-friendly. Thus, the TP4056 module is essential for safe battery charging and power management.

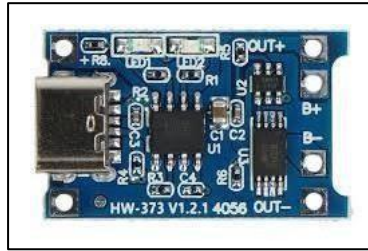


Fig 5.5 TP4056 MODULE

5.6 BREAD BOARD

A breadboard is a reusable prototyping tool used for building and testing electronic circuits without the need for soldering. It consists of a rectangular plastic board with a grid of holes internally connected by metal strips, allowing electronic components such as resistors, capacitors, sensors, and microcontrollers to be easily inserted and interconnected. The breadboard is typically divided into terminal strips and power rails, where the terminal strips are used for placing components and the power rails are used to distribute supply voltage and ground throughout the circuit. This arrangement enables quick modifications, making it ideal for experimentation, learning, and debugging circuit designs. Breadboards are widely used in educational settings, research, and project development, including IoT and embedded systems like smart insoles, where temporary circuit assembly and testing are essential before final implementation.

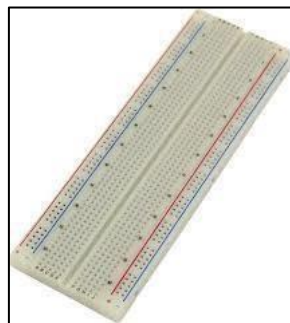


Fig 5.6 BREAD BOARD

5.7 CONNECTING WIRES

Connecting wires are used to establish electrical connections between different components of the circuit. They carry signals and power from one part of the system to another. Proper wiring is essential for correct functioning of the circuit. High-quality wires ensure minimal signal loss and reliable connections. They are used extensively during both prototyping and final implementation. Wires help in maintaining proper communication between Arduino, sensors, displays, and other modules

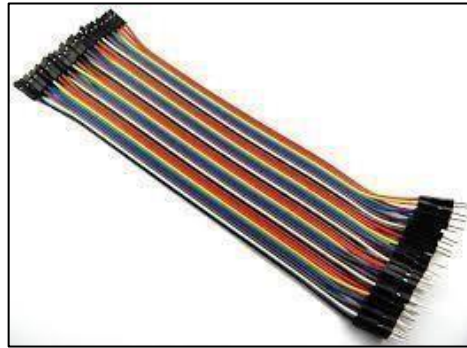


Fig 5.7 CONNECTING WIRES

CHAPTER 6

SOFTWARE TOOL

6. SOFTWARE TOOL

6.1 Arduino IDE

Arduino IDE is an open-source programming environment used to write, edit, compile, and upload code to Arduino-compatible boards. It supports Windows, Linux, and macOS, providing a simple graphical interface for embedded development in C/C++. The IDE organizes code into sketches with two main functions: setup and loop. The setup function runs once at startup to initialize variables, configure input/output pins, and set up serial communication. The loop function then runs repeatedly, allowing the system to continuously read sensors (such as the smart insole's pressure sensors) and control outputs (for example, sending data or activating indicators).

Before uploading the code, you must select the correct board type (for example, Arduino Nano or an ESP32 module) and the correct serial (COM) port in the Tools menu. Then click Verify to compile the sketch and check for syntax errors. If the code complies successfully, click Upload to transfer the machine code to the insole's microcontroller via a USB cable. The Arduino IDE uses a built-in compiler (and uploader like AVRDUDE for AVR-based boards) to convert the sketch into a hexadecimal file stored in the device's memory. It also provides a Serial Monitor for real-time debugging and data display, showing information sent from the smart insole to the computer.

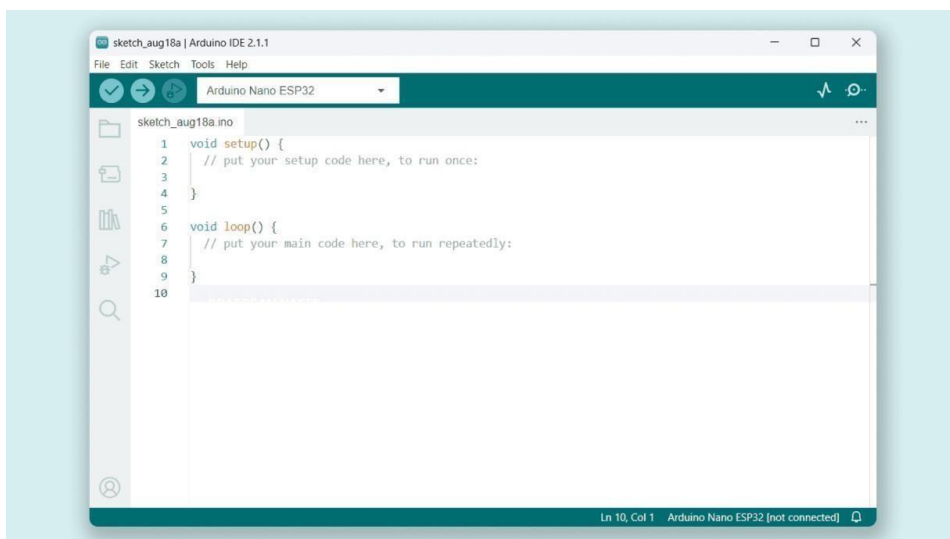


Fig 6.1: Arduino IDE

Key feature of the Arduino IDE includes numerous built-in libraries for interfacing with hardware components (e.g. sensors, communication modules, LCDs) and simplifying common tasks. These libraries speed up development by providing pre-written code for devices like pressure sensors or Bluetooth modules. In the Smart Insole project, we use the Arduino IDE to develop separate sketches for the insole's microcontroller (transmitter) and any receiver unit. For example, the insole's sketch reads foot pressure and motion data and transmits this data (via Wi-Fi, Bluetooth, or another interface), while the receiver sketch (on a smartphone app or a secondary display unit) processes incoming data and shows the results on an LCD or mobile app. This makes the Arduino IDE essential for implementing the complete Smart Insole system.

6.2 Blynk IoT Application

Blynk IoT is a powerful cloud-based platform designed for developing Internet of Things (IoT) applications with ease. It provides a user-friendly mobile interface and cloud services that enable real-time monitoring and control of hardware devices. In the Smart Insole project, Blynk IoT plays a crucial role as the main interface between the user and the system, allowing continuous tracking of foot pressure distribution, motion activity, and step count.

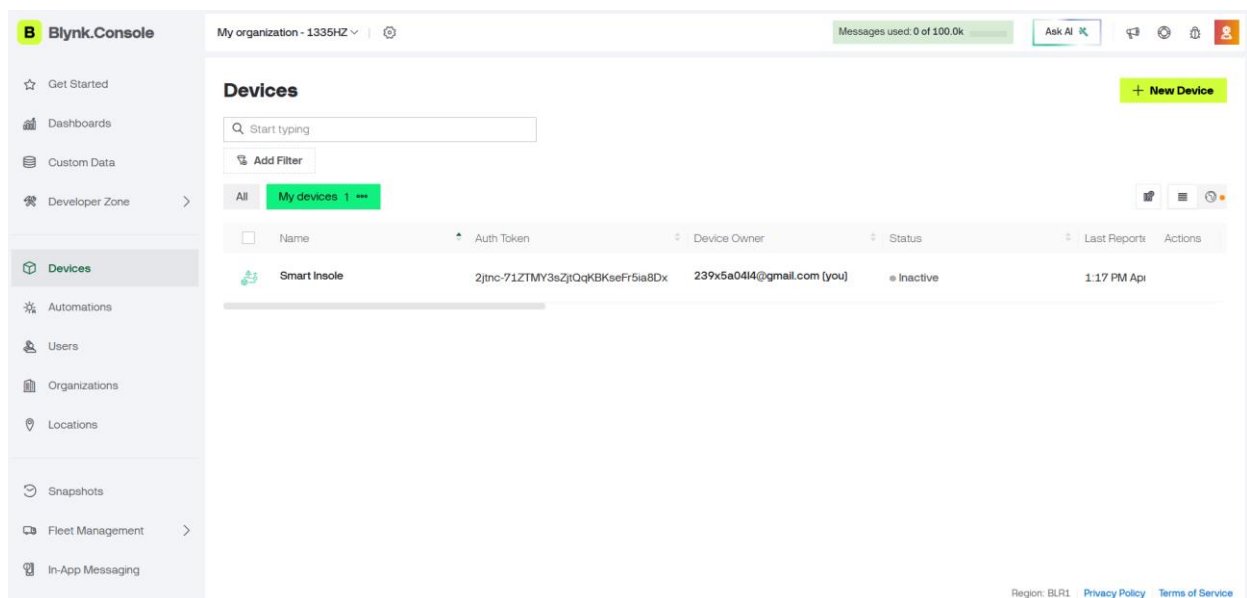


Fig 6.2.1: Blynk IOT Website Interface

The process of setting up a Blynk IoT account for the Smart Insole project begins at the official Blynk website, where you must first register using a valid email address to establish your organization's digital workspace. Once logged in, you navigate to the Templates section to create a blueprint for the hardware, specifically selecting the ESP32 as the controller and WIFI as the primary connection method. The next critical step involves defining DataStream, where you manually link the sensor parameters from your code such as pressure, temperature, and step count—to specific Virtual Pins (V1 through V11) that act as the communication channels between the insole and the cloud.

After the DataStream are established, you transition to the Web Dashboard tab to design the user interface by dragging and dropping widgets like gauges for pressure distribution, value displays for clinical reports, and switches to trigger the session timer. To finalize the link between the website and your physical device, you must go to the Search menu to add a new device from your template, which generates a unique Authorization Token. This token, along with the Template ID and Name, must be copied directly into the header of your Arduino code, enabling the ESP32 to securely transmit real time gait data and receive control commands from the web interface.

The ESP32 microcontroller is connected to the internet via Wi-Fi and communicates with the Blynk cloud server using an authentication token. Sensor data collected from components such as pressure sensors and motion sensors is processed by the ESP32 and transmitted to the Blynk platform in real time. The mobile application displays this data using various widgets such as gauges, super charts, value displays, and LED indicators, providing a clear and interactive visualization of the user's foot health status.

The Blynk dashboard can be customized according to project requirements. For example, pressure values can be displayed using gauge widgets, motion or activity data can be shown using graphs, and abnormal conditions such as uneven pressure distribution can trigger notification alerts. This helps in early detection of issues like imbalance, improper walking patterns, or foot stress, which is especially beneficial for elderly users.

During operation, when the device is online, the dashboard updates continuously with live sensor readings, ensuring accurate monitoring. In case the device goes offline, the dashboard stops updating and may show the last received values along with an offline status indication. This helps the user identify connectivity issues instantly.



Fig 6.2.2: Blynk IoT Application interface

Blynk IoT also supports features like event notifications, data logging, and remote access. Users can receive alerts on their smartphones if sensor values exceed predefined thresholds. Additionally, historical data can be analyzed using graph widgets for better understanding of walking patterns over time. In this Smart Insole project, Blynk IoT eliminates the need for wired communication or external display units by providing a wireless, efficient, and easy-to-use monitoring solution. Thus, it significantly enhances the functionality, usability, and real-time performance of the overall system.

CHAPTER 7

RESULTS

RESULTS

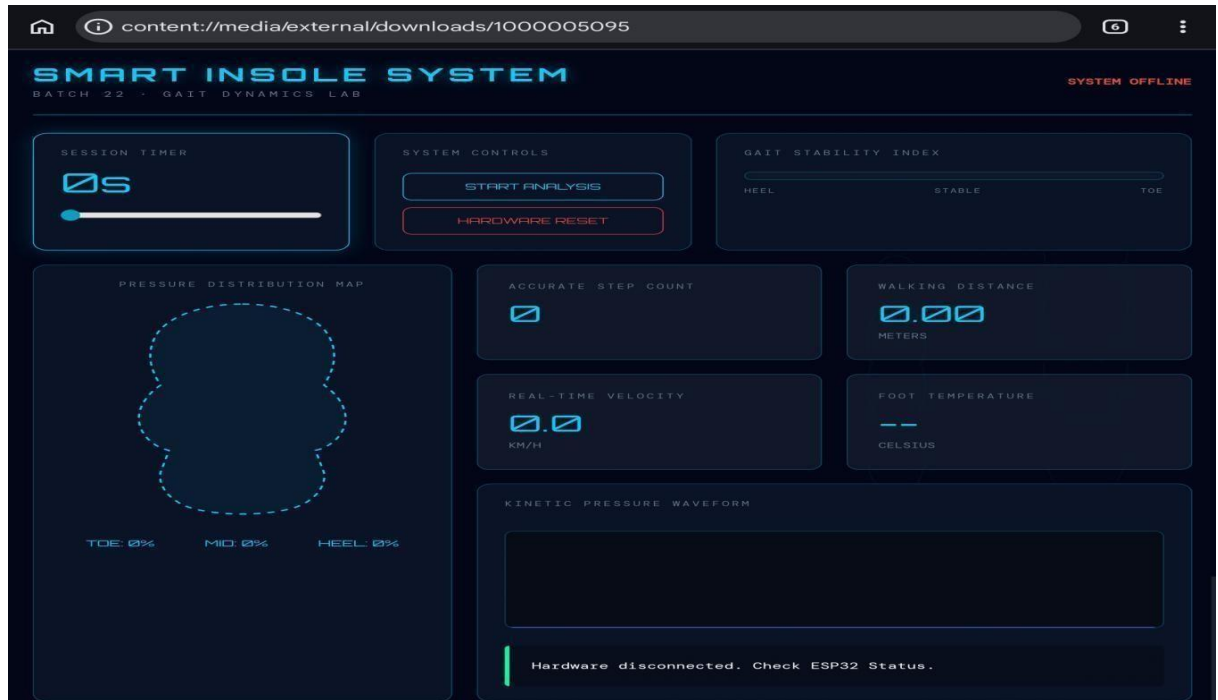


Fig 7.1 Blynk IoT Dashboard (Template)

The above figure shows Displays the dashboard status when the ESP32 device is offline, indicating no real- time data updates and showing connection loss or last recorded values.



Fig 7.2 Blynk IoT Dashboard (Device Online State)

The above figure Displays real-time sensor data from the ESP32 when the device is online,

with active updates of pressure and motion values on the dashboard.



Fig 7.3: Output Display in Blynk IoT Application

The above figure Displays real-time sensor data from the Smart Insole using ESP32, including pressure and motion values, through gauges and graphs for remote monitoring.

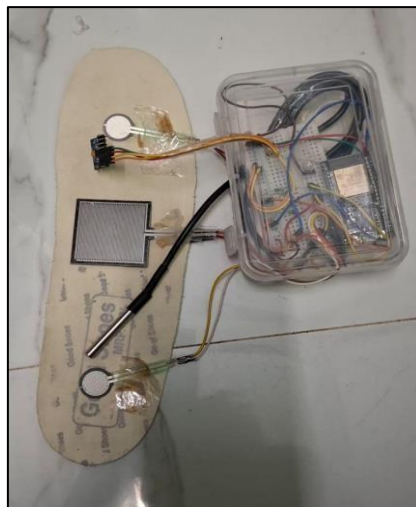


Fig 7.4: Interior look of Smart Insole System

The figure Displays the internal structure of the smart insole system by using ESP32.



Fig 7.5 Exterior look of Smart Insole System

The above figure Displays Exterior look of smart insole system which shows the final product of the shoe of insole system.

CHAPTER 8

ADVANTAGES & APPLICATIONS

ADVANTAGES AND APPLICATIONS

ADVANTAGES

- **Real-Time Monitoring:** Continuously tracks foot pressure, gait patterns, and temperature, providing instant feedback to the user.
- **Early Detection of Problems:** Helps in identifying abnormalities such as uneven pressure distribution, improper walking patterns, and potential foot disorders at an early stage.
- **Non-Invasive and Comfortable:** Designed to fit inside regular shoes, allowing users to monitor their health without affecting daily activities.
- **Portable and Wearable:** Compact and lightweight design makes it easy to use anytime and anywhere.
- **Wireless Communication:** Uses Bluetooth/Wi-Fi to transmit data, enabling remote monitoring through smartphones or computers.
- **Cost-Effective Solution:** Compared to traditional clinical equipment, the system is affordable and accessible.

APPLICATIONS

- **Healthcare Monitoring:** Used to monitor foot pressure, gait patterns, and temperature for early detection of foot related problems.
- **Elderly Care:** Helps in monitoring balance and movement of elderly individuals, reducing the risk of falls and improving safety.
- **Rehabilitation:** Assists patients recovering from injuries or surgeries by tracking their walking patterns and progress during recovery.
- **Sports and Fitness:** Used by athletes to analyze performance and optimize running or walking techniques.
- **Gait Analysis:** Helps in studying walking patterns for medical diagnosis and research purposes.
- **Remote Health Monitoring:** Enables doctors or caregivers to monitor patients remotely through wireless data transmission.
- **Smart Wearable Technology:** Can be integrated into IoT-based systems for advanced health tracking and smart lifestyle applications.

CHAPTER 9

CONCLUSION & FUTURE SCOPE

CONCLUSION AND FUTURE SCOPE

CONCLUSION

The Smart Insole System is an innovative and efficient solution for real-time monitoring of foot health and movement. By integrating sensors such as FSR, MPU6050, and temperature sensors with the ESP32 microcontroller, the system is capable of analyzing foot pressure distribution, gait patterns, and temperature variations accurately.

This project successfully demonstrates a portable, cost-effective, and user-friendly approach to continuous foot health monitoring. It helps in early detection of abnormalities, supports rehabilitation, and improves overall mobility and safety. The wireless communication feature enables real-time data transmission and easy accessibility for users and healthcare providers.

Overall, the Smart Insole System bridges the gap between traditional clinical monitoring and daily health tracking, making it a valuable application in healthcare, sports, and wearable technology.

FUTURE SCOPE

The Smart Insole System can be further improved by integrating other sensors to monitor additional health parameters and enhance accuracy. The use of machine learning and artificial intelligence can enable automatic detection of gait abnormalities and early prediction of health issues. The system can be connected to mobile applications and cloud platforms for real-time monitoring, data storage, and remote access by healthcare professionals.

Future developments may also focus on improving battery life, reducing power consumption, and designing a more flexible and comfortable insole. Additionally, the system can be expanded for applications in sports performance analysis, rehabilitation, and advanced healthcare monitoring systems.

REFERENCES

- [1] S. Patel et al., “A review of wearable sensors and systems for health monitoring,” IEEE, 2012. <https://www.mdpi.com/1424-8220/21/8/2821/>
- [2] M. Shu et al., “Smart insole system for gait analysis,” Sensors Journal, 2019. <https://www.researchgate.net/publication/310424686>
- [3] A. Raj et al., “IoT-based health monitoring systems,” 2021. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8073163/>
- [4] J. Smith et al., “Foot pressure monitoring using smart insoles,” 2020. <https://www.sciencedirect.com/science/article/pii/S2352648322000368>
- [5] K. Lee et al., “Wearable healthcare devices,” 2022. <https://pubmed.ncbi.nlm.nih.gov/40238890/>
- [6] P. Kumar et al., “Smart wearable systems for medical applications,” 2023. <https://pubmed.ncbi.nlm.nih.gov/31817067/>

